

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
2	(a)	(i)		[Minimum] energy needed to eject an electron [from a material or surface or metal not atom]	1			1		
		(ii)	I	Brighter light of same frequency means more photons [per second] but individual photons unchanged (accept hf unchanged) (1) [so] ejection rate increased (1) accept more electrons emitted but maximum KE unchanged (1)	3			3		
			II	Photon energy for frequency 7.4×10^{14} Hz is 4.9×10^{-19} J or by implication (1) $\phi = 3.7 \times 10^{-19}$ [J] (1) Photon energy for frequency 5.1×10^{14} Hz is 3.4×10^{-19} [J] or threshold frequency = 5.59×10^{14} Hz (1) or by implication ecf on arithmetic slip on ϕ Photon energy < work function or photon frequency < threshold frequency <u>so no emission</u> or equivalent (1) no ecf Statement that KE is negative treat as neutral			4	4	3	
	(b)	(i)		Photon energy expressed as $E = \frac{hc}{\lambda}$ or by implication (1) Number per second = $\frac{P\lambda}{hc}$ (1) Accept $P \div (\frac{hc}{\lambda})$ or equivalent		2		2	2	
		(ii)		Use of $p = \frac{h}{\lambda}$ seen in momentum change per sec = $N \times \frac{h}{\lambda}$ ecf (1) Simplified to $\frac{P}{c}$ ecf from (i) but not if left in terms of N (1) First mark available even if no answer to (b)(i)		2		2	2	
		(iii)		Force	1			1		
				Question 2 total	5	4	4	13	7	0